A SAI MOULI

Backend Software Engineer

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PROFESSIONAL SUMMARY

Software engineer at NatWest Group building backend APIs and data pipelines for an observability platform and an AI analytics engine over a 136-table database. Builds policy-gated agent systems on FastAPI/PostgreSQL; merged fixes into FastMCP and DSPy.

EXPERIENCE

Software Engineer | NatWest Group Bengaluru, India | Jul 2025 – Present

- Developed backend APIs and data pipelines for the **SCO Observability Platform**, letting internal banking teams monitor applications and track customer journeys end to end across interconnected systems.
- Built data processing for application lifecycle events and technical observability data, giving analytics and quality-control teams visibility into application status and workflow progress to investigate issues.
- Optimized backend APIs and data-processing workflows, cutting large dashboard queries from **~3 minutes to under 30 seconds**.
- Built an AI natural-language analytics engine over a **136-table** reporting database: a text-to-SQL workflow generates SQL from plain English, validates it against the schema, executes it and charts the result.
- Integrated the engine into the dashboard as a chatbot for self-service reporting, replacing ad-hoc query requests to engineers.

Backend Developer Intern | Aamara Technologies Bengaluru, India | Mar 2024 – Aug 2024

- Built the backend of Techno Rings' shop-floor operations system, used by **20,000 people**, on Django REST Framework and MySQL with a React front end: machine assignment, shift logging and tool-life tracking.
- Implemented payroll exception paths as model-level rules: breakdowns pro-rate shift and target, and workers move to another machine's job. Daily processing fell from **about four hours to under thirty minutes**.

Software Development Intern | Pods Studio Bengaluru, India | Nov 2023 – Dec 2023

- Moved customer email onto event-driven Celery/cron triggers (retention up **27%**); debugged and fixed failed in-app payments.

OPEN-SOURCE CONTRIBUTIONS

PrefectHQ/fastmcp 2 PRs merged

- **PR #4758 (merged):** fixed a lost-update race in the event store (a five-concurrent-write reproduction kept 1 of 5 event IDs); after maintainer review, replaced a store-wide lock with **64 striped asyncio locks** keyed by stream ID.
- **PR #4705 (merged):** stopped the response cache storing is_error results, which pinned a transient 503 for a full hour.

stanfordnlp/dspy | **PR #10720 (merged):** fixed dspy.majority() raising TypeError when voting on non-string values 1 PR merged

chroma-core/chroma | **Issue #7560:** showed CI lint could never fail; **PR #7559 (open)** fixes two hidden bugs 1 issue, 1 open PR

567-labs/instructor | **PR #2846 (open):** renders context placeholders in input_text parts that reached the model raw 1 open PR

PROJECTS

Zippyy | Python, FastAPI, PostgreSQL, Redis, Next.js github.com/SaiMouli3/zippyy

- Built a multi-tenant delivery-recovery platform: signed carrier, WhatsApp and voice webhooks are de-duplicated into an inbox, drive a shipment state machine, and wake durable agent workflows that act only through typed tools.
- Every carrier action must pass a deterministic 14-rule policy engine, and the LLM reads only replies the rule extractor is unsure of. Per-merchant row-level security, SKIP LOCKED job leases; 145 tests pass.

XTXT | Go, Python, JavaScript, Rust, Java, C github.com/SaiMouli3/xtxt

- Designed a plain-text format whose directives carry images, tables, charts and typed records as JSON-extractable data.
- Wrote the spec, Go reference parser and a 12-case conformance suite run in CI by 5 SDKs; published to PyPI and npm.

TECHNICAL SKILLS

Languages: Python, C++, SQL
Backend Engineering: FastAPI, Django, Django REST Framework, Flask, SQLAlchemy, REST APIs, data pipelines, WebSockets, Celery
AI Engineering: LangGraph, LangChain, MCP servers, RAG pipelines, embeddings, FAISS
Databases & Infrastructure: PostgreSQL, MySQL, Redis, AWS, Docker, Kubernetes, Linux, Git, GitHub Actions
Engineering Practices: asyncio concurrency, database transactions, pytest, API design, performance optimization

EDUCATION

Ramaiah Institute of Technology | B.Tech, Computer Science and Engineering | CGPA: 9.06/10 2021 – 2025